LECTURE 4

RECAP OF LAST LECTURE

Tensor product of two quantum states.

$$|\phi\rangle = \begin{bmatrix} \alpha_0 \\ \vdots \\ \alpha_{N-1} \end{bmatrix}, \quad |\psi\rangle = \begin{bmatrix} \beta_0 \\ \vdots \\ \beta_{M-1} \end{bmatrix}$$

$$|\phi\rangle \otimes |\psi\rangle = \begin{bmatrix} \alpha_0 \\ \vdots \\ \alpha_{N-1} \end{bmatrix} \otimes \begin{bmatrix} \beta_0 \\ \vdots \\ \beta_{M-1} \end{bmatrix} = \begin{bmatrix} \alpha_0 \beta_0 \\ \alpha_0 \beta_1 \\ \vdots \\ \alpha_0 \beta_{M-1} \\ \alpha_1 \beta_1 \\ \vdots \\ \alpha_{N-1} \beta_{M-1} \end{bmatrix} \in \mathbb{C}^{MN}$$

NOTE: $\sum_{i=0}^{N-1} \sum_{j=0}^{M-1} |\alpha_i \beta_j|^2 = \sum_{i=0}^{N-1} |\alpha_i|^2 \sum_{j=0}^{M-1} |\beta_j|^2 = 1$

Example

$$|\phi\rangle = \begin{bmatrix} \alpha_0 \\ \alpha_1 \end{bmatrix} \qquad |\psi\rangle = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}$$

qubit

$$|\phi\rangle \otimes |\psi\rangle = \begin{bmatrix} \alpha_0 \beta_0 \\ \alpha_0 \beta_1 \\ \alpha_1 \beta_0 \\ \alpha_1 \beta_1 \end{bmatrix} = \alpha_0 \beta_0 |0\rangle \otimes |0\rangle + \alpha_0 \beta_1 |0\rangle \otimes |1\rangle + \alpha_1 \beta_1 |1\rangle \otimes |0\rangle + \alpha_1 \beta_1 |1\rangle \otimes |1\rangle$$

$$= \alpha_0 \beta_0 |00\rangle + \alpha_0 \beta_1 |01\rangle + \alpha_1 \beta_0 |10\rangle + \alpha_1 \beta_1 |11\rangle$$

Partial measurement: (creates a mixed state)

Let $|\phi\rangle = \begin{bmatrix} \alpha_{00} \\ \alpha_{01} \\ \alpha_{10} \\ \alpha_{11} \end{bmatrix}$

Suppose we just measure 2^{nd} qubit.

- w.p. $|\alpha_{00}|^2 + |\alpha_{10}|^2$, the state collapses to $\dfrac{\alpha_{00}|00\rangle + \alpha_{10}|10\rangle}{\sqrt{|\alpha_{00}|^2 + |\alpha_{10}|^2}}$

- w.p. $|\alpha_{01}|^2 + |\alpha_{11}|^2$, the state collapses to $\dfrac{\alpha_{01}|01\rangle + \alpha_{11}|11\rangle}{\sqrt{|\alpha_{01}|^2 + |\alpha_{11}|^2}}$

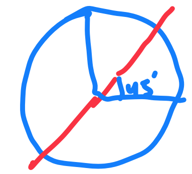
Some frequently used unitary transformations.

$\boxed{\text{NOT}}$

$|0\rangle \rightarrow |1\rangle$

$|1\rangle \rightarrow |0\rangle$

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

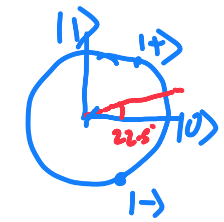


reflection

$\boxed{H}$

$$|0\rangle \rightarrow |+\rangle = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$$

$$\begin{bmatrix} \frac{1}{\sqrt{2}} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix}$$



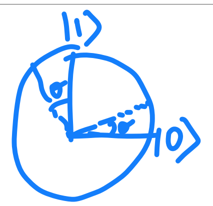
$$|1\rangle \rightarrow |-\rangle = \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix}$$

Reflection : self inverse

$\boxed{S_\theta}$

$$|0\rangle \rightarrow \begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix}$$

$$|1\rangle \rightarrow \begin{bmatrix} -\sin\theta \\ \cos\theta \end{bmatrix}$$



$\boxed{\text{CNOT}}$

NOT 2nd qubit if 1st is 1.

$|00\rangle \rightarrow |00\rangle$

$|01\rangle \rightarrow |01\rangle$

$|10\rangle \rightarrow |11\rangle$

$|11\rangle \rightarrow |10\rangle$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$\boxed{\text{CSWAP}}$

SWAP 2nd/3rd qubit if 1st is 1.

$|000\rangle \rightarrow |000\rangle$

$|001\rangle \rightarrow |001\rangle$

$|010\rangle \rightarrow |010\rangle$

$|011\rangle \rightarrow |011\rangle$

$|100\rangle \rightarrow |100\rangle$

$|101\rangle \rightarrow |110\rangle$

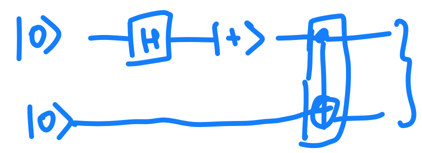
$|110\rangle \rightarrow |101\rangle$

$|111\rangle \rightarrow |111\rangle$

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Control operations introduce entanglement.

Creating Bell state



$$\frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle = \begin{bmatrix} 1/\sqrt{2} \\ 0 \\ 0 \\ 1/\sqrt{2} \end{bmatrix}$$

Entangled state : cannot be written as

$$|\phi\rangle \otimes |\psi\rangle = \begin{bmatrix} \alpha_0 \beta_0 \\ \alpha_0 \beta_1 \\ \alpha_1 \beta_0 \\ \alpha_1 \beta_1 \end{bmatrix}$$

Similarities with probability theory:

- probability mass $\rightsquigarrow$ amplitude
- conditioning on a random variable taking a certain value $\rightsquigarrow$ measurement. (committing to a state)
- Independent random variables $\rightsquigarrow$ unentangled state $|\phi\rangle \otimes |\psi\rangle$
- Dependent/corelated random variables $\rightsquigarrow$ entangled state.

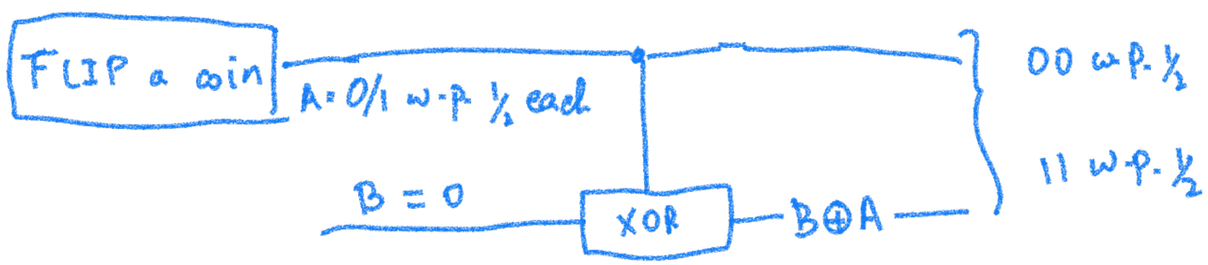
Difference from prob. theory:

- quantum state is in multiple classical states "at the same time"
- Unitary transform allows to "introduce randomness"

  e.g. $|0\rangle \longrightarrow \boxed{H} \longrightarrow |+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$
- Amplitudes can be negative (allows for cancellation)

  e.g. $|+\rangle - \boxed{H} - |0\rangle$

A classical process analogous to creating Bell State



TODAY:

- NO CLONING THEOREM
- QUANTUM TELEPORTATION
- ARE QUANTUM COMPUTERS AT LEAST AS POWERFUL AS CLASSICAL

Q.M. Laws needed for Computation:

(1) Initialize a quantum state $|0\rangle$ (w.l.o.g.)

(2) Unitary transformation

(3) (Partial) Measurement. (in standard basis w.l.o.g.)

NO CLONING THEOREM (Wootters Zurek 82)

There is no physical device that can create a copy of an unknown quantum state

(2 unentangled copies of $|\psi\rangle$)

for all qubits $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

CLONER does not exist!

- The proof is very simple: just uses linearity
- One can make a similar statement for coin flips.

WHAT WE CAN BUILD

$\boxed{|0\rangle \text{ maker}} \rightarrow |0\rangle\ |0\rangle \ldots$

$\boxed{|+\rangle \text{ maker}} \rightarrow |+\rangle\ |+\rangle\ |+\rangle \ldots$

For any given α, β

$\boxed{\alpha|0\rangle + \beta|1\rangle \text{ maker}} \rightarrow |\psi\rangle\ |\psi\rangle \ldots$

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

What is impossible

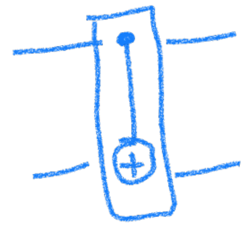
$$|\psi\rangle \longrightarrow \boxed{\text{CLONER}} \longrightarrow |\psi\rangle\,|\psi\rangle \;\ldots\ldots$$

ATTEMPT AT BUILDING CLONER

$\alpha|0\rangle + \beta|1\rangle$

$\otimes$

$|0\rangle$



CNOT

$$\alpha|00\rangle + \beta|11\rangle$$

$$= \begin{bmatrix} \alpha \\ 0 \\ 0 \\ \beta \end{bmatrix}$$

$$|\psi\rangle \otimes |\psi\rangle = \begin{bmatrix} \alpha \\ \beta \end{bmatrix} \otimes \begin{bmatrix} \alpha \\ \beta \end{bmatrix}$$

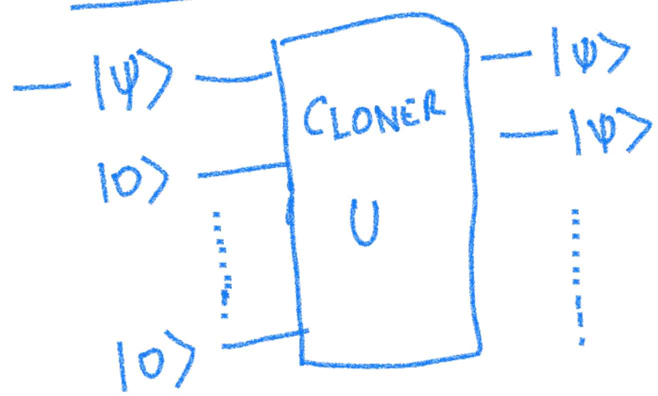
$$\alpha|0\rangle + \beta|1\rangle \, , \, \alpha|0\rangle + \beta|1\rangle$$

$$\alpha^2|00\rangle + \alpha\beta|01\rangle + \beta\alpha|10\rangle + \beta^2|11\rangle = \begin{pmatrix} \alpha^2 \\ \alpha\beta \\ \alpha\beta \\ \beta^2 \end{pmatrix}$$

We want $\begin{bmatrix} \alpha \\ \beta \end{bmatrix} \otimes \begin{bmatrix} \alpha \\ \beta \end{bmatrix} = \begin{bmatrix} \alpha^2 \\ \alpha\beta \\ \alpha\beta \\ \beta^2 \end{bmatrix}$

Why is this impossible via any unitary transformation?

Proof: Suppose there exists a CLONER



$$U\left(|\psi\rangle \otimes |0 \ldots 0\rangle\right) = |\psi\rangle \otimes |\psi\rangle \otimes |\ldots\ldots\rangle$$

$$U\left(|0\rangle \otimes |0\ldots\ldots 0\rangle\right) = |0\rangle \otimes |0\rangle \otimes |f(0)\rangle$$

$$U\left(|1\rangle \otimes |0\ldots\ldots 0\rangle\right) = |1\rangle \otimes |1\rangle \otimes |f(1)\rangle$$

$$U\left(|+\rangle \otimes |0\ldots\ldots 0\rangle\right) = |+\rangle \otimes |+\rangle \otimes |f(+)\rangle$$

$$\underbrace{\frac{1}{\sqrt{2}} U\left(\underline{|0\rangle} \otimes |0\ldots\ldots 0\rangle\right) + \frac{1}{\sqrt{2}} U\left(\underline{|1\rangle} \otimes |0\ldots 0\rangle\right)}$$

$$= U\left(|+\rangle \otimes |0\ldots\ldots 0\rangle\right)$$

$$\frac{1}{\sqrt{2}} |0\rangle \otimes |0\rangle \otimes |f(0)\rangle + \frac{1}{\sqrt{2}} |1\rangle \otimes |1\rangle \otimes |f(1)\rangle$$

$$= |+\rangle \otimes |+\rangle \otimes |f(+)\rangle$$

$$|+\rangle \otimes |+\rangle = \frac{1}{2}\left(|00\rangle + |01\rangle + |10\rangle + |11\rangle\right)$$

Measure 1st 2 qubits. Contradiction

Classical analogy?

- How would you clone a biased coin of unknown bias that you can toss only once?

Quantum Teleportation

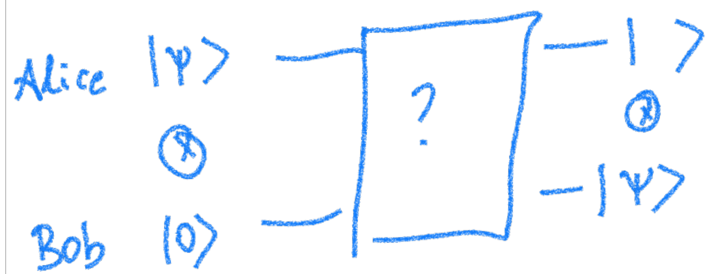
- Alice received this qubit $|\psi\rangle$ as a present
- She wants to send it to her friend Bob who lives far away.

Alice

Bob

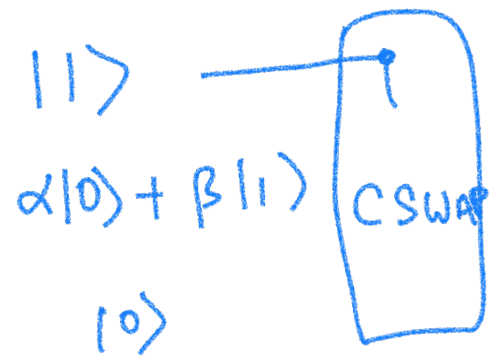
PUZZLE

- Assume they can do basic unitary operations
- Alice cannot go to Bob and give him the qubit
  (text message is ok)



→ We want these unentangled, so Alice cannot destroy present by measurement.

What can they do?

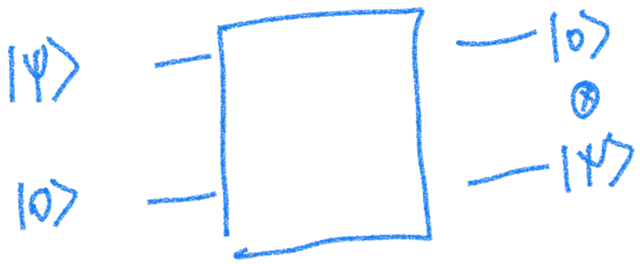


$$\alpha|100\rangle + \beta|110\rangle \to \alpha|100\rangle + \beta|101\rangle$$

$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$,

Let U be such that $U|0\rangle = |\psi\rangle$

e.g. $U = \begin{bmatrix} \alpha & \beta^* \\ \beta & -\alpha^* \end{bmatrix}$

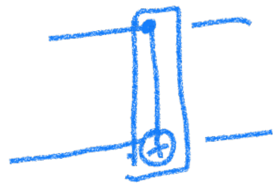


Problem?

Alternative idea

$\alpha|0\rangle + \beta|1\rangle$

$|0\rangle$



CNOT

$\alpha|00\rangle + \beta|11\rangle$

$\downarrow \boxed{H}$

$\alpha|+0\rangle + \beta|-1\rangle$

$$\frac{\alpha}{\sqrt{2}}|00\rangle + \frac{\alpha}{\sqrt{2}}|10\rangle + \frac{\beta}{\sqrt{2}}|01\rangle - \frac{\beta}{\sqrt{2}}|11\rangle$$

On measurement Alice sees

$b = 0$ w.p. $\frac{|\alpha|^2 + |\beta|^2}{2}$ and state collapses to

$\alpha|00\rangle + \beta|01\rangle$

$b = 1$ w.p. $\frac{1}{2}$ and state collapses to

$\alpha|10\rangle - \beta|11\rangle$

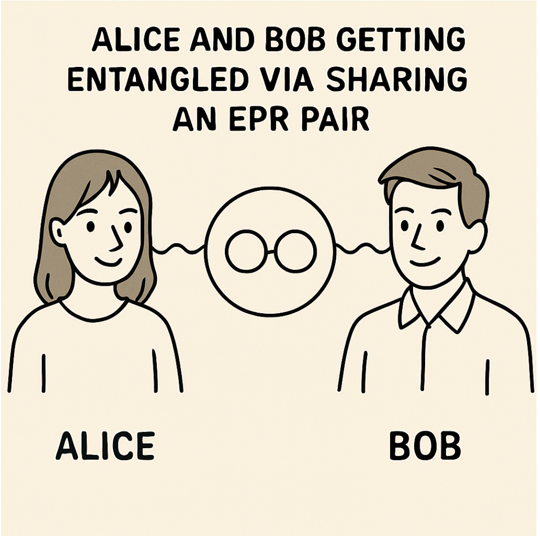
Alice sends b to Bob

If $b = 0$ then Bob has $|\psi\rangle$

If $b = 1$ then Bob applies

$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$ to get $|\psi\rangle$

Before Bob left for his expedition



$$\frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$$

Their current joint state

Alice

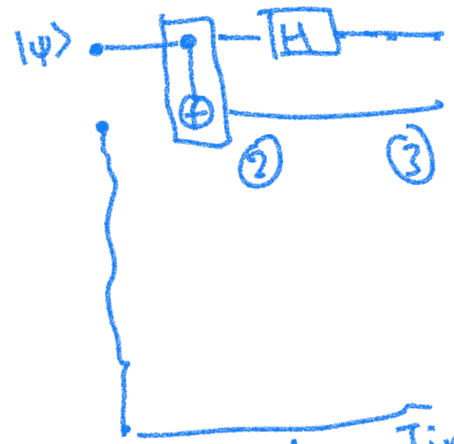
Bob

$$|\psi\rangle \otimes \frac{1}{\sqrt{2}}|00\rangle + \frac{1}{\sqrt{2}}|11\rangle$$

$= \alpha|0\rangle + \beta|1\rangle$

Joint state: $\frac{\alpha}{\sqrt{2}}|000\rangle + \frac{\alpha}{\sqrt{2}}|011\rangle + \frac{\beta}{\sqrt{2}}|100\rangle + \frac{\beta}{\sqrt{2}}|111\rangle$

Can Alice use this to locally implement the previous computation



$$\frac{\alpha}{\sqrt{2}}|000\rangle + \frac{\alpha}{\sqrt{2}}|011\rangle + \beta/\sqrt{2}\,|100\rangle + \beta/\sqrt{2}\,|111\rangle$$

Time 2:

$$\frac{\alpha}{\sqrt{2}}|000\rangle + \frac{\alpha}{\sqrt{2}}|011\rangle + \beta/\sqrt{2}\,|110\rangle + \beta/\sqrt{2}\,|101\rangle$$

Time 3

$$\frac{\alpha}{2}|000\rangle + \frac{\alpha}{2}|100\rangle + \frac{\alpha}{2}|011\rangle + \frac{\alpha}{2}|111\rangle + \beta/2\,|010\rangle - \beta/2\,|110\rangle + \frac{\beta}{2}|001\rangle - \beta/2\,|101\rangle$$

① If Alice measures 00, state collapses to

$$\alpha |000\rangle + \beta |001\rangle$$

Bob's state is $\alpha|0\rangle + \beta|1\rangle \because |\psi\rangle$

② Alice measures 01 then Bob's state is

$$\alpha|1\rangle + \beta|0\rangle$$

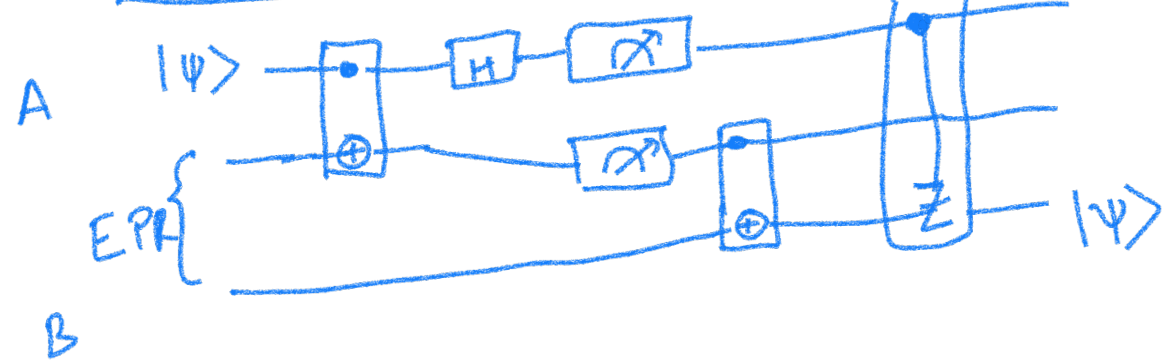
③ Alice measures 10 then Bob's state is

$$\alpha|0\rangle - \beta|1\rangle$$

Alice measures $|11\rangle$ implies Bob's state is

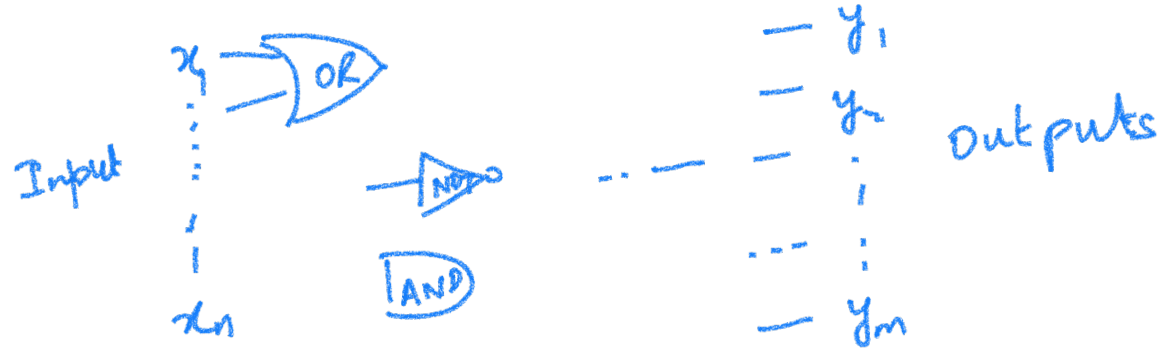
$$\alpha|1\rangle - \beta|0\rangle$$

Overall



Classical computing

Let $x_1, \ldots \ldots, x_n \in \{0,1\}$



Circuit computes $F : \{0,1\}^n \rightarrow \{0,1\}^m$

Example: F could take an n-bit number and output a factor of n.

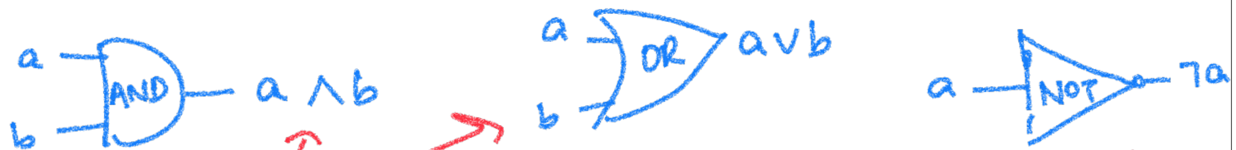
Efficiency: Measured by # of Gates as a function of n.

Gates $\approx$ Steps / Running time

Cook/Karp

Thm: Any algorithm that runs in time T and computes $f: \{0,1\}^n \to \{0,1\}^m$ can be converted into a boolean ckt. with $O(T \log T)$ AND/OR/NOT gates that computes f.

- How to build quantum circuits that implement classical circuits
- Quantum operations are reversible
- What about



Not Reversible

Reversible

CAN WE IMPLEMENT THESE VIA A REVERSIBLE QUANTUM CKT.

- Lets look at non-trivial quantum gates for which output qubits are "bits" when input qubits are "bits"

Assume $|a\rangle$, $|b\rangle$, $|c\rangle$ are either $|0\rangle$ or $|1\rangle$